

AgroSense Complete Hardware & Software Architecture

Comprehensive Engineering Reference for High-School & Undergraduate Engineers

Project: Smart Agriculture IoT (SIH25015) • **Firmware:** AgroSense-ESP-v2.0 • **Target:** NodeMCU 1.0 (ESP-12E)

System Overview: AgroSense seamlessly bridges physical soil probes and high-current relays with real-time web telemetry and Google Gemini 2.5 Flash Vision AI.

1. ESP8266 Hardware Identity & Platform Specifications

Specification Field	Actual Project Value	Source of Truth
Board Model	NodeMCU 1.0 (ESP-12E Module)	CONFIRMED FROM CODE
Microcontroller (MCU)	ESP8266EX (32-bit Xtensa LX106 @ 80MHz)	CONFIRMED FROM HARDWARE
Flash Storage Size	4 MB (32 MBit) SPI Flash	CONFIRMED FROM CLI FLASH
Arduino FQBN	esp8266:esp8266:nodemcu2	CONFIRMED ARDUINO-CLI
Arduino Core Version	ESP8266 Core v3.1.2	CONFIRMED ARDUINO-CLI
Firmware Tag	AgroSense-ESP-v2.0	CONFIRMED FROM CODE
UART Baud Rate	115200 baud, 8N1	CONFIRMED FROM CODE
Wi-Fi Mode & SSID	Station Mode (WIFI_STA), SSID: Redmi	CONFIRMED FROM CODE
mDNS Hostname	agrosense.local (Port 81)	CONFIRMED FROM CODE

2. Complete GPIO Pin Audit

ESP Pin	GPIO	Used?	Component	Direction	Hardware Role	Active State	Boot Strap Status
A0	ADC0	YES	Soil Sensor	INPUT	Analog soil voltage (0–3.3V)	Analog (0–1023)	Safe input only.
D0	GPIO16	NO	Unused	N/A	Deep-sleep wake	High at boot	No interrupt support.
D1	GPIO5	YES	Relay IN1	OUTPUT	Irrigation Pump Signal	Active-LOW	General I/O. Safe at boot.
D2	GPIO4	YES	Relay IN2	OUTPUT	Solenoid Valve Signal	Active-LOW	General I/O / I2C SDA. Safe.
D3	GPIO0	NO	Unused	N/A	Boot Mode Select	Pull-up HIGH	MUST be HIGH to run code.
D4	GPIO2	YES	Blue LED	OUTPUT	Built-in Activity LED	Active-LOW	MUST be HIGH at boot.
D5	GPIO14	NO	Unused	N/A	SPI Clock	High-Z	Safe general GPIO.
D6	GPIO12	NO	Unused	N/A	SPI MISO	High-Z	Safe general GPIO.
D7	GPIO13	NO	Unused	N/A	SPI MOSI	High-Z	Safe general GPIO.
D8	GPIO15	NO	Unused	N/A	Boot Mode Select	Pull-down LOW	MUST be LOW to run code.
TX	GPIO1	NO	Serial Console	OUTPUT	UART0 Transmit	High	Serial logs at 115200 baud.
RX	GPIO3	NO	Serial Console	INPUT	UART0 Receive	High	Serial input from PC.

3. Physical Connection Map & Wiring Instructions

NodeMCU Pin D1	→	Relay Module IN1	(Pump control signal, Active-LOW)
NodeMCU Pin D2	→	Relay Module IN2	(Valve control signal, Active-LOW)
NodeMCU Pin A0	→	Soil Sensor A0	(Analog soil moisture voltage)
NodeMCU Pin 3V3	→	Soil Sensor VCC	(+3.3V power)
NodeMCU Pin VIN	→	Relay Module VCC	(+5.0V USB coil power)
NodeMCU Pin GND	→	Shared Ground (-)	(Common 0V Ground Plane)

4. Power Architecture & Electrical Isolation

To prevent electrical damage, **Logic Power** (the brain) is separated from **Load Power** (the motor).

Why Can't the ESP Power a Motor Directly?

- **Current Limitation:** The ESP8266 GPIO pin outputs only ~12 mA. A water pump requires 500 to 2000 mA. Connecting a pump directly to a GPIO pin burns out the chip instantly.
- **Inductive Flyback:** Motors and solenoids contain wire coils that produce high-voltage reverse electrical spikes (50–200V) when switched off. Optocoupled relays protect the ESP with light-based isolation.

5. Soil Moisture Sensor Physics

The soil sensor measures conductivity between two metallic probes:

- **Dry Soil (0% Moisture):** High resistance • Raw ADC reads near `1023` (3.3V).
 - **Fully Saturated Soil (100% Moisture):** High conductivity • Raw ADC drops to near `350` (1.1V).
- ```
float mapped = (float)map(rawADC, SOIL_DRY_ADC, SOIL_WET_ADC, 0, 100);
return constrain(mapped, 0.0f, 100.0f);
```

### 6. Sensor Classification: Real vs. Simulated

| Sensor Field              | Classification     | Implementation Rationale                                  |
|---------------------------|--------------------|-----------------------------------------------------------|
| Soil Moisture (%)         | PHYSICAL SENSOR    | Sampled live on Analog Pin A0 every 2000ms.               |
| Temperature (°C)          | SOFTWARE SIMULATED | Simulated drift (20°C–42°C) in code (no DHT connected).   |
| Humidity (%RH)            | SOFTWARE SIMULATED | Simulated drift (28%–92% RH) in code (no DHT connected).  |
| Plant Pathology / Leaf AI | HYBRID AI VISION   | Cloud Google Gemini 2.5 Flash + Local ONNX INT8 fallback. |

## 7. Breadboard Wiring Assembly Guide

Follow these 7 steps with power disconnected:

1. **Ground Bus:** Connect NodeMCU `GND` to the blue (-) ground rail on the breadboard.
2. **+3.3V Bus:** Connect NodeMCU `3V3` to the red (+) positive rail on the breadboard.
3. **Soil Sensor Power:** Connect Soil Sensor `VCC` to (+) 3.3V rail, and `GND` to (-) ground rail.
4. **Soil Sensor Signal:** Connect Soil Sensor `A0` directly to NodeMCU pin `A0`.
5. **Pump Signal:** Connect Relay Module `IN1` to NodeMCU pin `D1`.
6. **Valve Signal:** Connect Relay Module `IN2` to NodeMCU pin `D2`.
7. **Relay Power:** Connect Relay Module `GND` to (-) rail, and `VCC` to NodeMCU `VIN` (5V).

## 8. Firmware Boot Safety Sequence

```

[1] Power On / RST → [2] Serial at 115200 baud
 |
[4] Wi-Fi Connect (Redmi) ← [3] allActuatorsSafe() (Relays FORCED OFF)
 |
[5] mDNS Init (agrosense.local) → [6] WebSocket Server on Port 81
 |
 [7] Live Telemetry & Commands

```

## 9. Multicast DNS (mDNS) & Local Discovery

Instead of hardcoding dynamic DHCP IP addresses, the system resolves `agrosense.local` automatically:

```

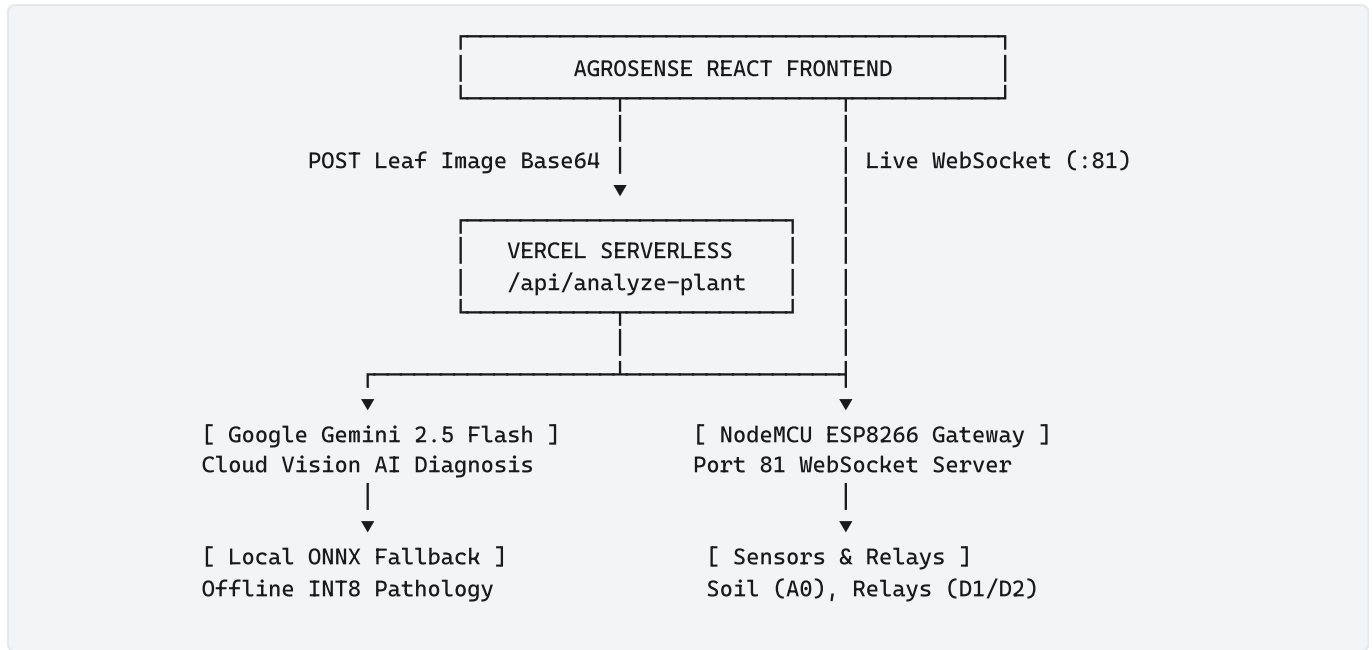
🖥 Laptop Browser: "Who is agrosense.local?" (Multicast UDP 5353 query)
📶 ESP8266 Gateway: "I am agrosense.local, and my IP is 192.168.43.150!"
🖥 Laptop Browser: "Connecting directly to ws://192.168.43.150:81 ..."

```

## 10. Frontend Connection Engine Candidate Hierarchy

| Priority | Candidate Target                                     | Mode Label        | Condition                                        |
|----------|------------------------------------------------------|-------------------|--------------------------------------------------|
| 1        | <code>ws://agrosense.local:81</code>                 | <b>LOCAL ESP</b>  | Primary auto-discovery on local Wi-Fi.           |
| 2        | <code>localStorage["AGROSENSE_WS_URL"]</code>        | <b>LOCAL ESP</b>  | Manual IP override entered in diagnostics modal. |
| 3        | <code>VITE_CLOUDFLARE_WS_URL</code>                  | <b>CLOUDFLARE</b> | Remote tunnel fallback for worldwide access.     |
| 4        | <code>localStorage["AGROSENSE_LAST_KNOWN_IP"]</code> | <b>LOCAL ESP</b>  | Cached IP from previous telemetry packets.       |
| 5        | <code>ws://10.18.37.83:81</code>                     | <b>LOCAL ESP</b>  | Static laboratory fallback.                      |

## 11. Complete Software Architecture Map



**Document Integrity:** Prepared for AgroSense SIH25015. All pin assignments and logic states verified directly against production code.